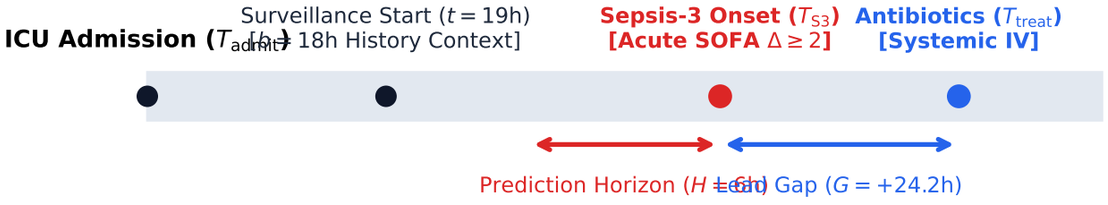
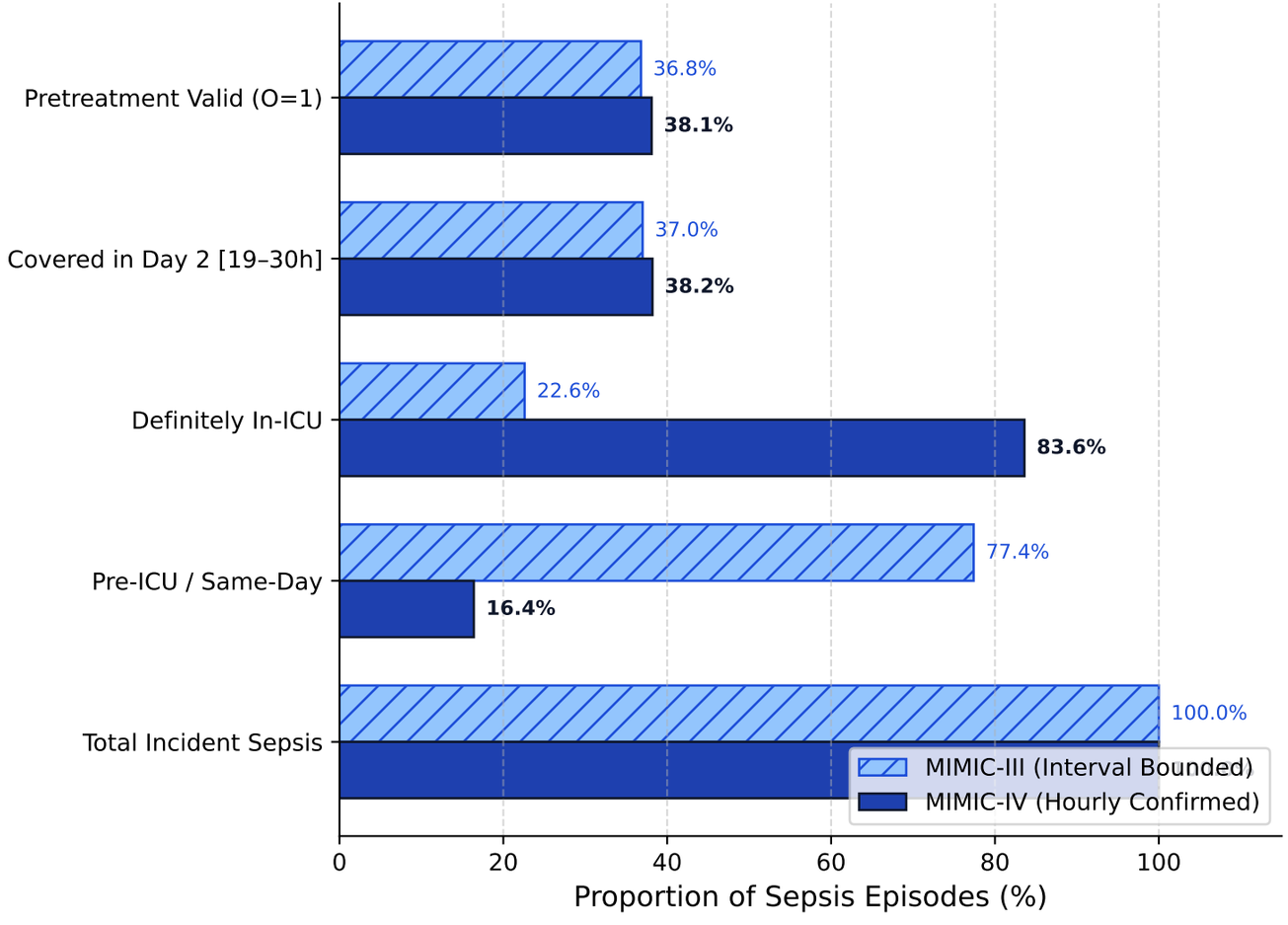
A. Operational Clocks & Timestamp Precision Hierarchy



Evidence Hierarchy for Clinical Treatment Timing:

Tier 1 (MIMIC-IV eMAR):
Confirmed administration with minute-level timestamps (Primary Actionability Benchmark)

Tier 2 (MIMIC-IV POE):
Timestamped computerized order entry (Order-based lead time)

Tier 3 (MIMIC-III Rx):
Calendar-date proxy [00:00, 24:00) with interval censoring (Bounded Sensitivity Proxy)

B. Opportunity Attrition & Interval Uncertainty